

Latency-Hiding Remote Entanglement with a Shuttled $^{40}\text{Ca}^+$ Ion Chain in a Glass Surface Trap

Experimental collaboration - draft author list and affiliations to be supplied

Abstract. We design a two-node trapped-ion link with 100 km total fibre path and midpoint heralding, using 2×10^4 $^{40}\text{Ca}^+$ ions in each glass surface trap. Beyond fibre attenuation, the dominant irreducible time cost is the 0.500 ms photon-and-herald propagation interval, limiting one emitter that waits after every attempt to 2000.000 attempts s^{-1} . Five latency-hiding architectures are compared, and temporal multiplexing by transporting a nine-ion chain through one fixed free-space interface is selected. A surface-electrode model extracted from the supplied STL predicts an RF null 41.307 μm above the trap, radial frequency 7.540013 MHz at 80 V amplitude and 39.15 MHz, and axial frequency 0.262280 MHz for the specified anisotropy. The equilibrium chain spans 58.205 μm . Invariant-based engineering produces a 100 μm , 10 μs single-ion move and an 88 μs nine-ion sequence. A sealed 100 kHz RC filter would leave 114.171 COM quanta after the nominal sequence; exact discrete pre-emphasis reduces every dwell-end prediction below 0.015 quanta. Two-PMT records give $g^{(2)}(0)=0.147\pm0.052$ with the best resolved 1.4 μs gate. Near-resonant red-sideband records give an apparent post-shuttling occupation of 3.043 ± 0.289 quanta. We propose a 397 nm free-space proof of concept followed by 854 nm photon generation and difference-frequency conversion to 1550 nm.

1. Introduction

Photonic heralding separates the local requirements of trapped-ion logic from the spatial scale of a network. A photon emitted by each node carries information about its parent ion; interference and a detector pattern at a midpoint station project the remote ions into an entangled state. The method has progressed from metre-scale demonstrations [1-3] to elementary-network rates with high fidelity [4] and trapped-ion links spanning buildings [5]. At 100 km, the central engineering difficulty is the product of small optical success probability and a classical round-trip constraint: an ion cannot safely be reset and reused until the herald for its time bin has returned.

For a total photon-plus-herald path of 100 km in fibre with group velocity $2.0 \times 10^8 \text{ m s}^{-1}$, the causal wait is 500 μs . Cooling, optical pumping, spontaneous emission, detector timing, and the 10 μs moves considered here are individually shorter. After excluding fibre attenuation as requested, propagation is therefore the dominant time budget for a wait-after-each-attempt emitter. Collection, coupling, conversion, detection, dark counts, and indistinguishability control success probability, but not the 2 kHz causal ceiling.

The available platform is well matched to temporal multiplexing: each node can hold a linear chain of up to ten ions, the optical system is fixed in free space, and ions can be moved. Multiple ions can therefore serve as a bank of communication memories. Each ion emits once, is translated away while retaining the photonic time-bin label, and waits coherently for its herald. This draft combines the network timing, the supplied trap mesh, shuttling control, photon statistics, ARTIQ Rabi records, and a low-loss wavelength plan.

This is a design and offline analysis of supplied apparatus records. It is not a claim that software autonomously operated high voltage, lasers, shutters, or vacuum hardware. Physical commissioning remains under laboratory interlocks and personnel control.

2. Architecture research and selection

2.1. Timing bottleneck

For two-photon midpoint heralding, let p_{gamma} include emission into the accepted mode, collection, single-mode coupling, conversion, fibre transmission, filtering, and detector efficiency. A representative linear-optics success probability is $p_{\text{link}} \sim (1/2)p_{\text{gamma,A}}p_{\text{gamma,B}}$. Loss controls p_{link} ; propagation controls emitter reuse. The one-emitter maximum is the inverse of $100 \text{ km} / (2.0 \times 10^8 \text{ m s}^{-1}) = 0.500 \text{ ms}$, or 2000.000 Hz.

The uphill/downhill geometry is not a material timing correction. The fractional gravitational shift $g \Delta h / c^2$ is about 1.1×10^{-16} per metre; even 1 km gives only about 82 Hz at 397 nm, negligible beside the $\text{Ca}^+ P_{1/2}$ linewidth. Fibre temperature and acoustic phase noise are much larger. A two-photon Hong-Ou-Mandel protocol is preferred initially because it relaxes first-order optical-phase stabilization compared with single-photon path entanglement, while still requiring active arrival-time and polarization matching.

2.2. Five candidate schemes

Scheme	Gain	Strength	Main risk	Stage
A. Shuttled temporal multiplexing	Up to 9 outstanding attempts	One beam and collector; existing chain; direct time-bin-to-ion map	Transport heating and optical crosstalk	Recommended
B. Parallel multi-focus interface	Up to 9 simultaneous modes	No movement during emission	AOD/SLM paths, many modes and detectors, matching phases	Medium
C. Communication plus memory ions	Emitter state is transferred to storage	Separates optical cycling from long-lived memory; enables sympathetic cooling	Extra species or encoding and high-fidelity local gates	Medium/long
D. Cavity or nanophotonic interface	Collection enhancement plus multiplexing	Purcell enhancement and a defined spatial mode	Charging, cavity lock, integration with glass trap	Long
E. Segmented repeater chain	Shorter elementary latency and lower loss	Best scaling at much longer distance	Additional nodes, memories, swapping and routing	Network

Scheme A is selected because it directly exploits the chain and fixed beam and needs no new in-vacuum component. Nine attempts occupy only 88 us, far inside the 500 us wait. The nine-ion bank supports 18 kHz aggregate causal throughput, although the requested one-emitter rate remains 2 kHz. The architecture follows the quantum charge-coupled-device concept [6] and fast, low-excitation transport demonstrations [7-9]. Schemes C and D are attractive upgrades after the free-space proof of concept.

3. Methods

3.1. Trap field and pseudopotential

The STL spans 4.5 mm transversely and 9.0 mm axially. Its central boundaries identify two 20 um rails at $x=-52.5$ to -32.5 um and $x=+32.5$ to $+52.5$ um. These rails were set to 80 V zero-to-peak RF amplitude and every other electrode to ground. Since the rails are millimetres long while the ion height is tens of micrometres, the central section was solved with the standard gapless-plane surface-electrode boundary solution [10,11]. For a strip $[a,b]$, $\Phi=(V/\pi)\{\text{atan}[(x-a)/y]-\text{atan}[(x-b)/y]\}$. The RF pseudopotential is $U_{ps}=q^2|E|^2/(4m\Omega_{RF}^2)$, and its Hessian gives the radial secular frequency.

Rail identity has an experimental cross-check: the predicted axial frequency is close to the approximately 258 kHz carrier-to-sideband separation in the final ARTIQ campaign. Assigning the adjacent wider metal regions as RF instead predicts approximately 112 kHz axial confinement, inconsistent with that scale. A final FEM should still include named CAD electrodes, finite gaps, dielectric glass, DC voltages, feeds, and measured stray fields; the quoted result is a centre-region analytic extraction from the supplied mesh.

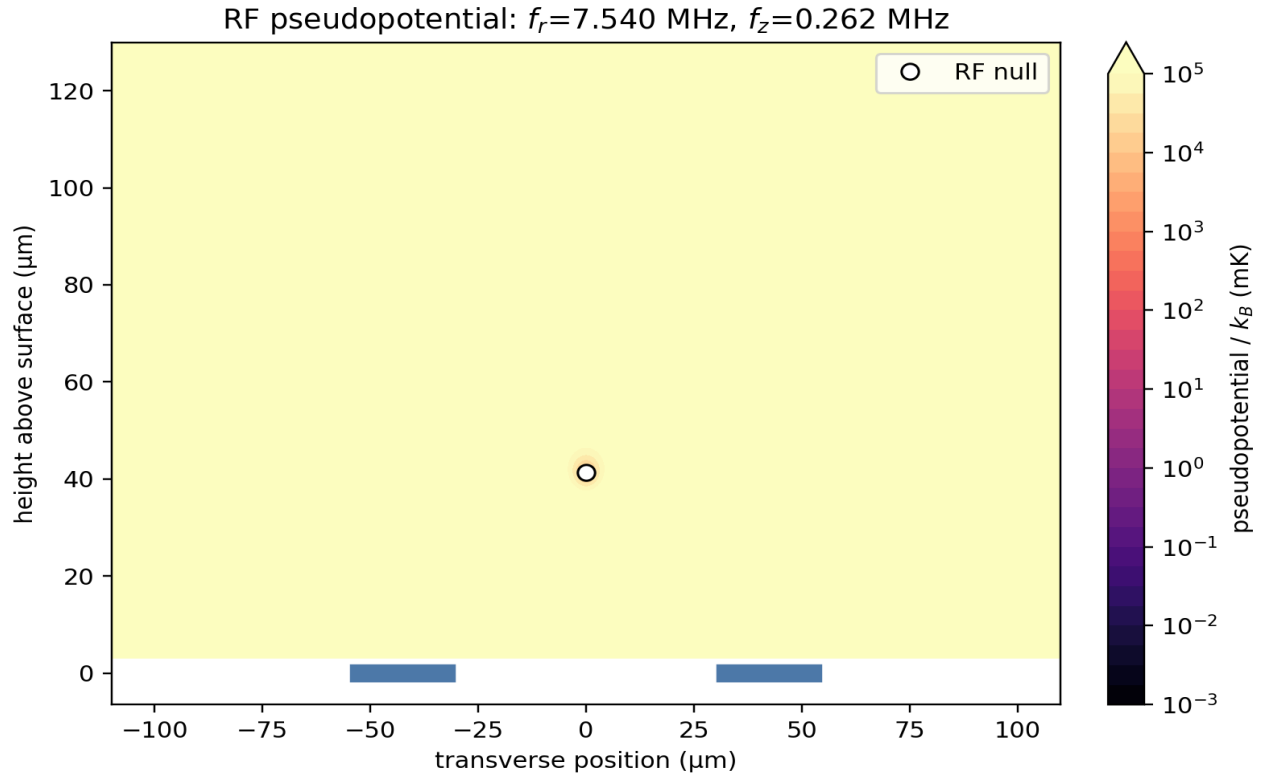


Figure 1. RF pseudopotential for the two central rails. The null is 41.307 μm above the surface; the inset resolves the harmonic region.

3.2. Nine-ion equilibrium

The supplied anisotropy $\alpha=(\omega_z/\omega_r)^2=0.00121$ sets $f_z=f_r\sqrt{\alpha}$. Equilibrium coordinates minimize harmonic and Coulomb energy using length scale $[e^2/(4\pi\epsilon_0 m \omega_z^2)]^{1/3}$ [12,13]. The COM frequency equals the single-ion axial frequency; excitation in quanta uses effective mass $9m$.

Move	Spacing (μm)	Cumulative (μm)
1	8.584027	8.584027
2	7.277415	15.861442
3	6.730960	22.592402
4	6.510300	29.102703
5	6.510300	35.613003
6	6.730960	42.343963
7	7.277415	49.621378
8	8.584027	58.205405

3.3. Invariant-engineered movement and RC compensation

For each harmonic move, $q_c(t)=q_i+d(10s^3-15s^4+6s^5)$, with $s=t/T$, has zero initial and final velocity and acceleration. The trap centre is $q_0=q_c+q\text{-}\ddot{q}_c/\omega_z^2$ [9], returning coherent displacement and momentum to zero at T . The single-ion file moves 100 μm in 10 μs . The chain file applies eight 10 μs moves at the calculated spacings, each followed by 1 μs dwell. Ion 1 begins at the beam and ions 2-9 are presented sequentially, ending at 88 μs .

The sealed filters obey $y\text{-}\dot{q}=2\pi f_c(u-y)$, $f_c=100$ kHz. The coupled state $[y,q,q\text{-}\dot{q}]$ was propagated exactly for every zero-order-held CSV interval. COM excitation is $n=M\omega_z\{(q-y)^2+(q\text{-}\dot{q}/\omega_z)^2\}/(2\hbar)$, $M=9m$. With $a=\exp(-\Delta t/\tau)$, the exact discrete inverse is $u_k=(y_{k+1}^*-ay_k^*)/(1-a)$. The first control sample was constrained to zero, so its small startup mismatch remains in the reported excitation.

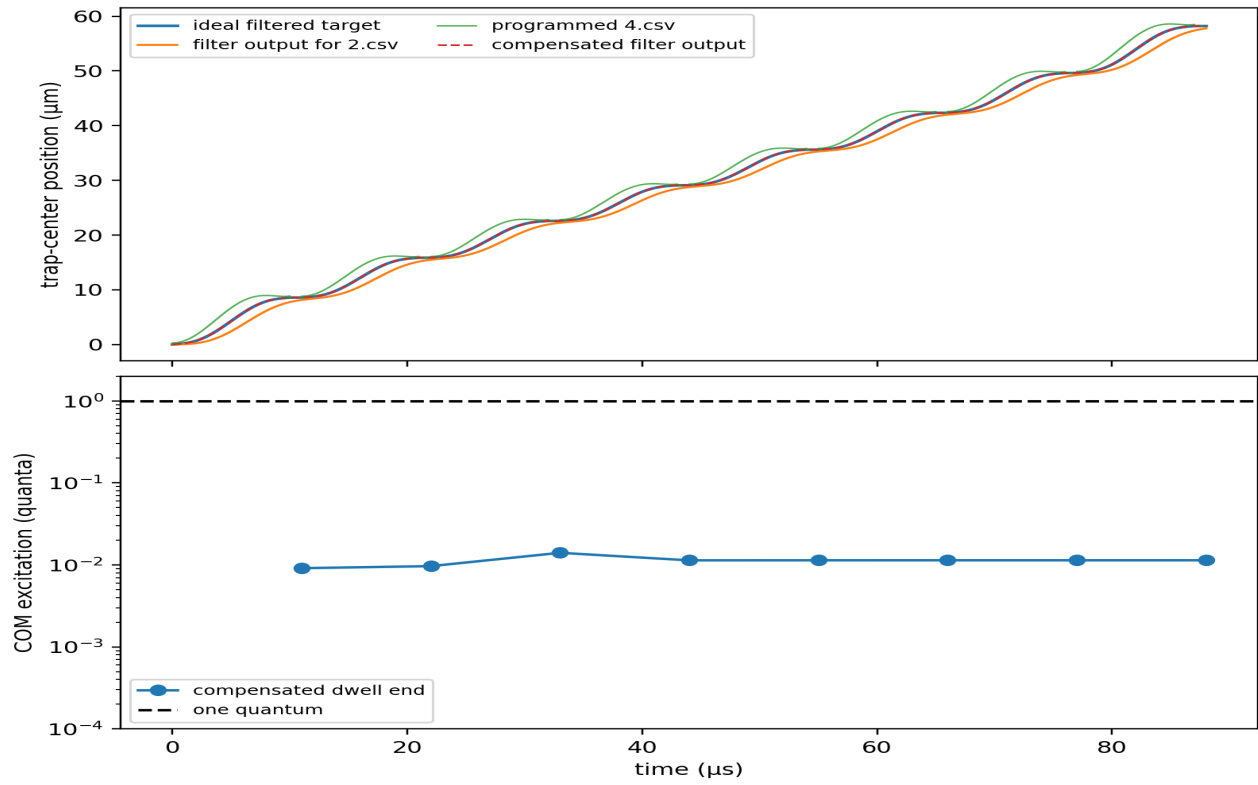


Figure 2. Ideal sequence, uncompensated filter response, programmed inverse-filter control, and compensated output. Every compensated dwell end is below one COM quantum.

3.4. 397 nm proof-of-concept source

The POC uses $4S_{1/2}$ to $4P_{1/2}$ excitation at 397 nm with 866 nm repumping from $3D_{3/2}$. The sequence is: Doppler cool the chain; optically pump the addressed ion; extinguish cooling and repump light at the detector; apply a short resonant excitation; accept spontaneous 397 nm light only after excitation leakage clears; and time tag two PMTs after a 50:50 beamsplitter. For ion-photon polarization entanglement, collect near the quantization axis and map σ^+/σ^- decay paths to $S_{1/2}$ Zeeman states.

A high-NA external objective should couple the ion into a filtered, polarization-maintaining single-mode path. Recommended leakage suppression is two cascaded AOMs, spatial separation, polarization rejection where selection rules allow, a narrow 397 nm interference filter, and detector acceptance after the excitation pulse. The recorded 1.7 μ s window is suitable for timing studies, but a future network attempt should be a single controlled excitation rather than unrestricted resonant scattering, which can create multiphoton probability.

3.5. Correlation and motional analysis

For each ion, the centre histogram contains unique zero-delay counts; the left and right files contain the ordered off-zero references and duplicate the centre. The centre file was therefore counted once at $n=0$. Side peaks $n=-10\dots-1$ and $1\dots10$ lie on an approximately 12 μ s comb. Gate widths were scanned from the native 0.2 μ s bin to the 1.7 μ s window. The requested 16 ns minimum is below the stored resolution, so no sub-bin interpolation was invented. Side-peak centres were calibrated using off-zero counts only. The lowest resolved value is tied at 1.4 and 1.6 μ s; 1.4 μ s is reported. Poisson error propagation gives the statistical uncertainty.

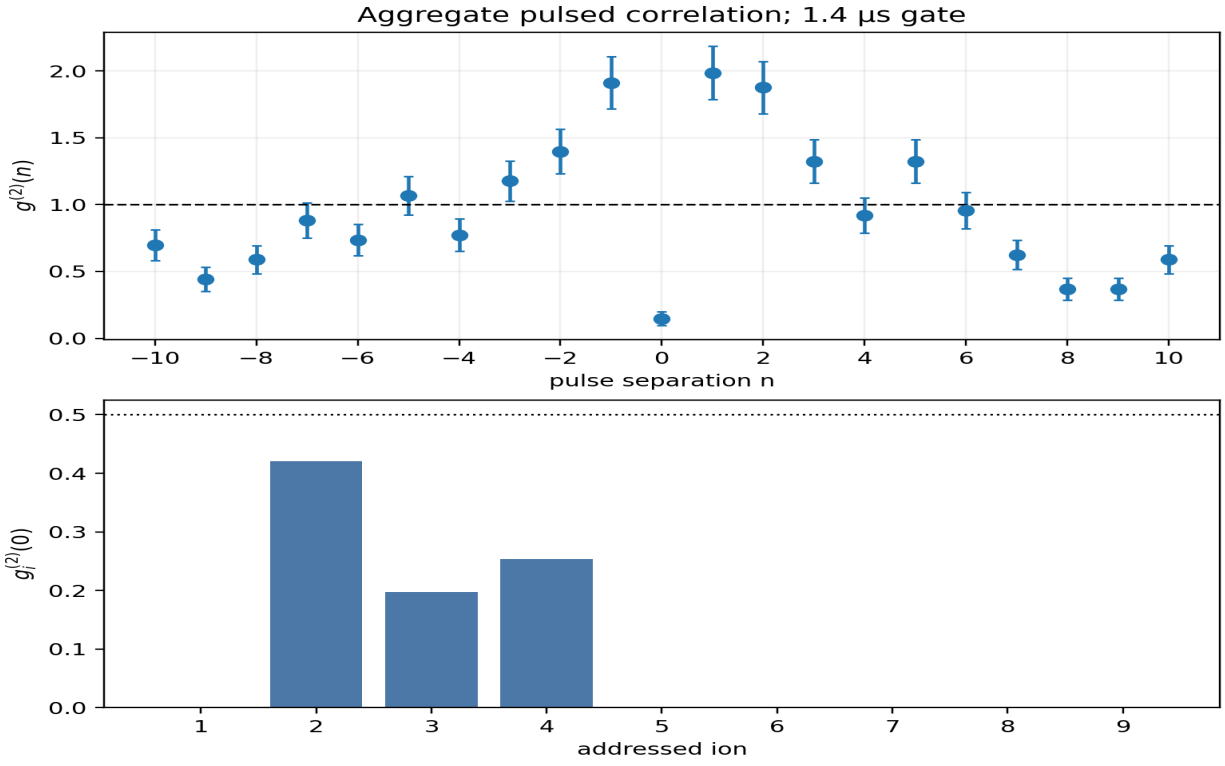


Figure 3. Aggregate pulsed $g^{(2)}(n)$ and per-ion zero-delay values. The central area is 8 counts and the accepted side-peak mean is 54.4 counts, giving $g^{(2)}(0)=0.147\pm0.052$.

Carrier run 9046 shows coherent Rabi oscillation. Near-resonant high-power red-COM-sideband runs 9050-9052 have pooled late-time dark probability $P_{\text{infinity}}=0.37633\pm0.00885$ for times at least 300 μs . For a resonant red sideband of a thermal mode after dephasing, $P_{\text{infinity}}=(1/2)\bar{n}/(1+\bar{n})$ for unit contrast and negligible background, giving $\bar{n}=3.043\pm0.289$. Detuning, imperfect transfer, and SPAM reduce the plateau, so this is an apparent occupation and approximately a lower bound. Publication-quality thermometry should interleave matched red and blue sidebands.

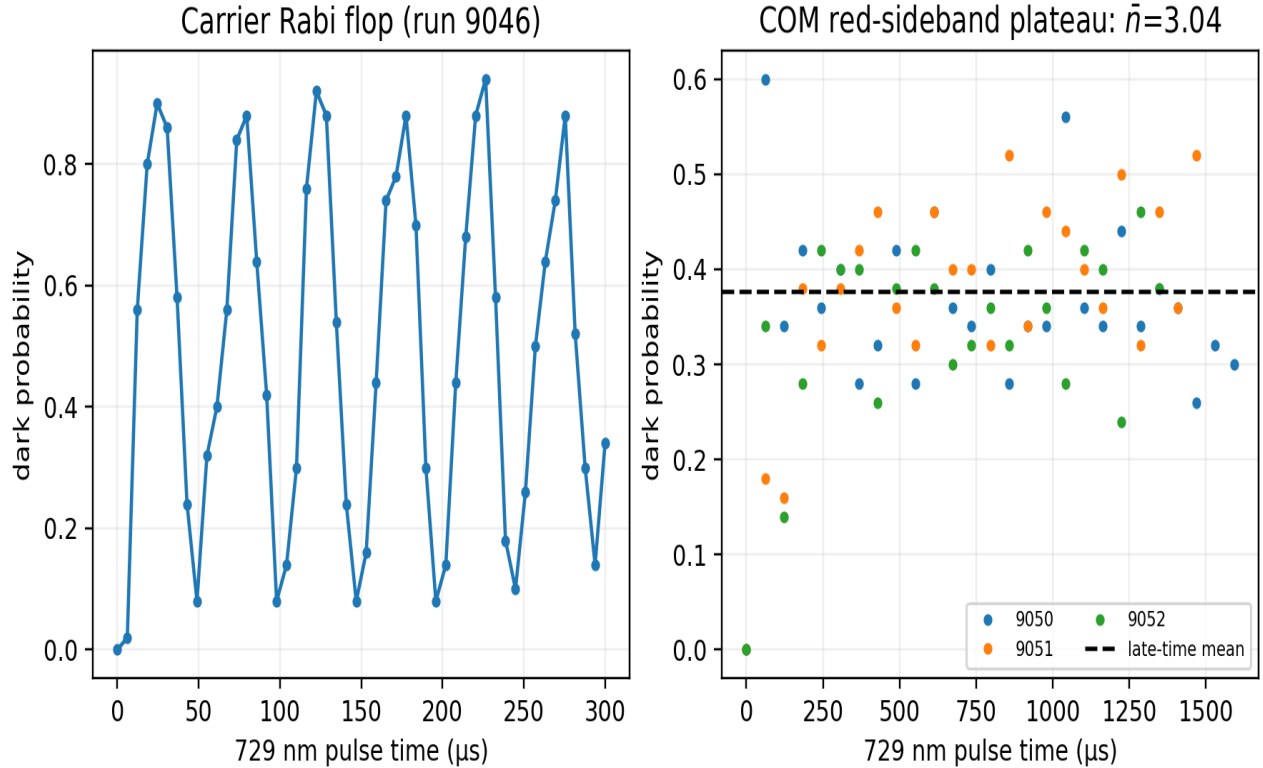


Figure 4. Supplied carrier flop and near-resonant red-sideband records. Coarse sideband sampling motivates the explicit plateau estimator rather than an underconstrained coherent thermal fit.

4. Results

4.1. Trap, chain, and transport

The RF null is 41.306779 μm high. The transverse field-gradient magnitude is $6.826 \times 10^9 \text{ V m}^{-2}$, giving $f_r = 7.540013 \text{ MHz}$ and $f_z = 0.262280 \text{ MHz}$. The chain is symmetric with outer ions at $\pm 29.103 \mu\text{m}$ and length 58.205405 μm . Spacings range from 6.510 μm at the centre to 8.584 μm at the ends.

The ideal trajectory is cold at move boundaries, but the fixed 100 kHz response leaves $n_{\text{COM,filtered}} = 114.170636$ at 88 μs . Even 20 μs moves leave 5.393 quanta because the 1 μs final dwell is shorter than the 1.592 μs filter time constant and phase-space displacement accumulates. Slowing alone does not meet the target.

Move (μs)	2	5	10	15	20
Final nCOM	6496.836	1038.361	114.171	20.954	5.393

The inverse-filter control spans 0.000 to 58.588 μm , only 0.382 μm beyond the final chain displacement. The largest dwell-end prediction is 0.014030 quanta and the final value is 0.011371. The design therefore meets the below-one-quantum requirement with large modeled margin. It must be commissioned move by move and updated with measured voltage-to-position transfer functions.

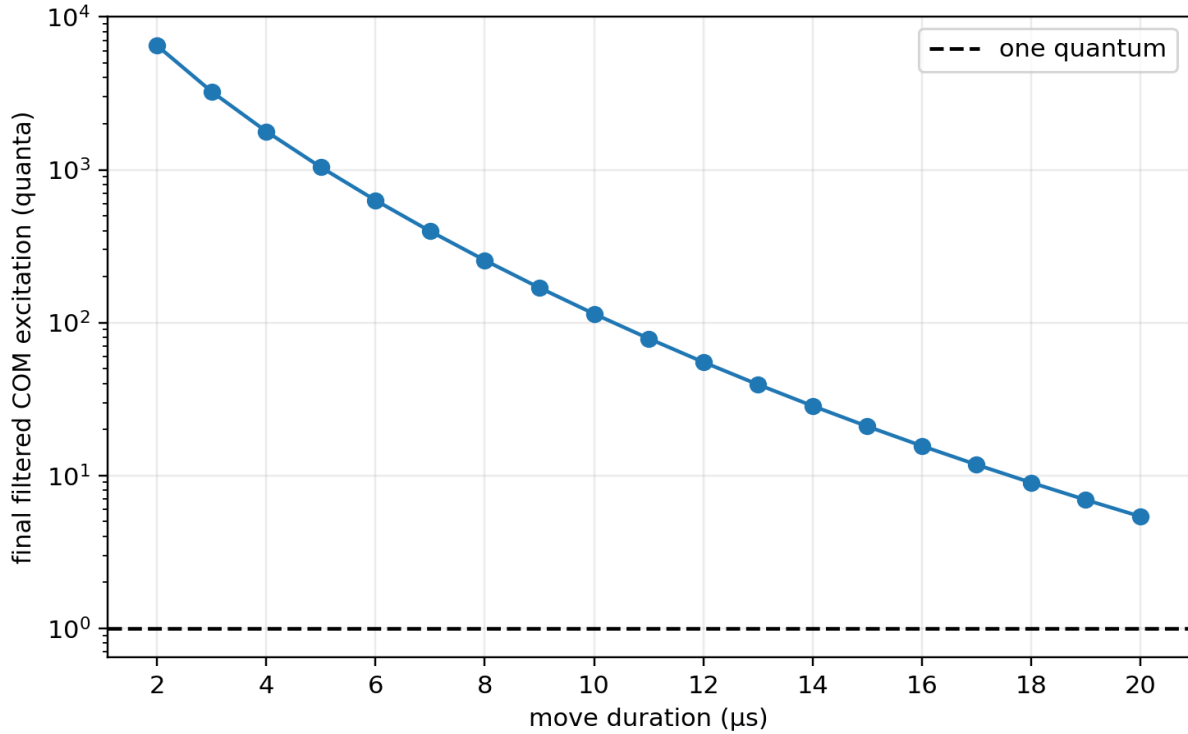


Figure 5. Final excitation after the 100 kHz filter for 2-20 μs uncorrected moves. None reaches the sub-quantum target.

4.2. Throughput and recorded experiment

One emitter waiting for the 100 km total path is limited to $\text{attempt_rate_hz} = 2000.000$. Nine labeled attempts fit in 88 μs , before any herald returns. The memory bank therefore supports $9 \times 2000 = 18,000 \text{ attempts s}^{-1}$ in steady state. The local value 9/88 μs is not the sustainable rate because only nine outstanding-memory slots exist. Failed ions are reset only after their time-bin results return; successful ions are moved to protected storage or the sequence pauses for network use.

The 397 nm data give $g^{(2)}(0) = 0.147059$, below the single-photon benchmark 0.5 but count limited. The apparent $\text{nbar} = 3.043 \pm 0.289$ is above the design target, consistent with historical uncompensated or imperfect transport. The delivered 4.csv is a prediction to test, not a claim that the archived experiment already used it.

5. Low-loss wavelength strategy

Direct 397 nm transport is unsuitable for 100 km silica fibre. The recommended network photon is in the telecom C band near 1550 nm, where modern fibre is typically about 0.18-0.20 dB km⁻¹. The mature Ca⁺ route is cavity-assisted Raman generation of an 854 nm photon on the 4P_{3/2} to 3D_{5/2} branch, followed by difference-frequency conversion in periodically poled lithium niobate. The relation $1/\lambda_{\text{pump}} = 1/854 \text{ nm} - 1/1550 \text{ nm}$ gives a pump near 1902 nm. Ion-telecom entanglement and tens-of-kilometres transmission have been demonstrated with this approach [16-18].

Direct 397 to 1550 nm difference-frequency conversion would require a pump near 534 nm. It is energetically possible but less attractive because of blue coupling, visible-pump noise, photorefractive effects, and polarization-preserving conversion. The 854 to 1550 nm route separates the POC from deployment and should use polarization-diverse conversion, active phase control, narrow filtering, and superconducting nanowire detectors. At 0.2 dB km⁻¹, 100 km gives about 20 dB total loss; each photon in a midpoint protocol travels about 50 km, or about 10 dB, before component loss.

6. Discussion

The analysis isolates two bottlenecks. At network scale, the 500 us causal delay dominates local sequence time. At control scale, the sealed filters, not the inverse-engineered harmonic path, dominate excitation. Both are solved by buffering: multiple ions buffer outstanding network attempts, and pre-emphasis buffers the electrode response. The 88 us chain sequence remains comfortably faster than the herald wait, leaving room for calibration or added settling without reducing the one-bank causal ceiling.

The recommended sequence is conservative: commission one 397 nm ion and characterize leakage, collection, coupling, and $g^{(2)}(0)$; load nine ions and test 4.csv one move at a time with matched red/blue thermometry; run the full 88 us local HBT source; duplicate timing at the second node and interfere locally; deploy fibre; then add 854 nm cavity emission and 1550 nm conversion. This order prevents source impurity, movement heating, indistinguishability, and fibre effects from masking one another.

Scheme C, communication ions plus protected memories, is the natural successor when more than nine outstanding attempts are needed. A mixed-species chain can support sympathetic cooling without resonant scattering from Ca⁺ memories. A cavity becomes compelling when collection rather than latency dominates. Neither upgrade is required for the present free-space POC.

6.1. Limitations

The STL is one watertight assembly rather than named electrode solids; RF rails were inferred from geometry and the sideband cross-check. The gapless model neglects dielectric glass, finite rail length, fabrication tolerances, RF pickup, and stray fields. DC confinement enters only through the supplied anisotropy. The CSV positions still require a multi-electrode voltage synthesis that preserves depth, radial-axis orientation, voltage limits, and slew limits.

The filter model assumes one identical first-order pole and zero-order-held samples. Real channels can have unequal gains, extra poles, saturation, and cross-coupling. The full transfer matrix should be measured with low-amplitude chirps and inverted with regularization. The $g^{(2)}$ files are histograms rather than raw time tags, so the 16 ns request cannot be applied below 0.2 us native bins. The Rabi data lack matched blue sidebands. These limitations should remain in a submission unless the missing calibrations and raw data are added.

7. Conclusion

A shuttled nine-ion temporal multiplexer is the most promising near-term architecture for this apparatus. It hides most of the 100 km herald latency without changing in-vacuum hardware and maps naturally to time-tagged midpoint heralding. The geometry predicts 7.540 MHz radial and 0.262 MHz axial confinement with a 58.205 um chain. The nominal waveform is unusable through the fixed filters because it leaves 114 COM quanta; the delivered inverse-filter control predicts below 0.015 quanta at every dwell. Existing records support 397 nm antibunching but show motion above target. The immediate experiment is a compensated nine-ion local HBT run, followed by midpoint interference and, for the long link, 854-to-1550 nm conversion.

Data and waveform availability

The accompanying numerical files are 1.csv, 2.csv, 3.csv, 4.csv, result.md, and analysis_summary.json. They record the 10,000-point grids, physical constants, equilibrium positions, filter prediction, and derived metrics used in this draft.

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